
EDUCATION

Tokyo, Japan	Institute of Science Tokyo	Spring 2024 – present
- Ph.D. in Information and Communications, March 2027 (expected). GPA: 3.86 - Research interest: Reinforcement learning, signal processing, optimization		
Tokyo, Japan	Tokyo Institute of Technology	Spring 2019 – March 2024
- M.S.E. in Information and Communications, March 2024. GPA: 3.85 - B.S.E. in Information and Communications, March 2022. GPA: 3.54		

ACADEMIC EXPERIENCE**Selected Publications**

- Y. Akiyama, **M. Vu**, K. Slavakis. **Nonparametric Bellman mappings for value iteration in distributed reinforcement learning**. Accepted to IEEE Transactions on Signal Processing, 2026.
- M. Vu**, K. Slavakis. **Gaussian-mixture-model Q-functions for Policy iteration in Reinforcement learning**. arXiv: 2512.18763, 2025.
- M. Vu**, K. Slavakis. **Online reinforcement learning via sparse Gaussian mixture model Q-functions**. arXiv: 2509.14585, 2025.
- M. Vu**, K. Slavakis. **Riemannian Q-function for policy iteration in reinforcement learning**. In European Signal Processing Conference (EUSIPCO), pp. 1672-1676, Palermo, Italy, September 8-12, 2025. (**nominated for Best Student Paper Award**)
- Y. Akiyama, **M. Vu**, K. Slavakis. **Nonparametric Bellman mappings for reinforcement learning: Application to robust adaptive filtering**. IEEE Transactions on Signal Processing, vol. 72, pp. 5644-5658, 2024.
- M. Vu**, Y. Akiyama, K. Slavakis. **Dynamic selection of p-norm in linear adaptive filtering via online kernel-based reinforcement learning**. In IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 1-5, Rhodes Island, Greece, June 4-10, 2023.

Reviewing

- 2026 IEEE International Conference on Acoustic, Speech and Signal Processing (ICASSP)**, 2025.
- IEEE Transactions on Signal Processing**, 2025.

Invited Talks

- An introduction to reinforcement learning**. FPT Japan x VJAI Meetup, Tokyo, April 2025.

ADDITIONAL EXPERIENCE AND AWARDS

- EUSIPCO Travel Grant**, European Association for Signal Processing (2025)
- Deputy Representative**: Awarded to students with outstanding academic achievements (2022)
- Japanese Government (MEXT) Scholarship** (2018 – present)

EMPLOYMENT

Software Engineer, Intern AI: Brushmind	Alphalogy	December 2024 – May 2025
- Developed and designed additional functions for AI-based photo editing. - Reduced rendering time in object removal function by 90% by implementing pretrained GAN models. - Conducted research and implemented stable diffusion models for mobile applications using Fast API.		
Teaching Assistant	Tokyo Institute of Technology	2024 – present
- Courses at Center for Data Science and Artificial Intelligence Education - Supported 100 students from different disciplines and graded weekly assignment submissions.		
Data Analyst, Intern	FPT Software Japan	January 2023 – June 2023
Applied statistic models to analyze factors which affect pH of water in an irrigation system.		

LANGUAGES AND TECHNOLOGIES

- Programming: Python; Julia; Matlab; Bash
- Tools: PyTorch; Manifolds.jl; Vim; Visual Studio; Git; Ubuntu/Unix